Network Scanner - AD Domain Mapper

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About

This script automates domain scanning, enumeration and exploitation stages for Active Directory networks. It combines tools such as nmap, masscan, CrackMapExec, Impacket utilities and hashcat to perform progressive levels of reconnaissance and attack.

Open-source tools used

nmap: TCP service discovery, NSE scripts, vulnerability scanning

masscan: High-speed UDP scanning (advanced)

CrackMapExec (CME): SMB/LDAP enumeration and password spraying

Impacket (GetNPUsers.py, samrdump.py): Kerberos and SAM enumeration

hashcat: Cracking extracted hashes/tickets (if available)

pandoc / wkhtmltopdf: PDF generation from Markdown fallback

Functionality overview

1. Getting the User Input

The script prompts the user for the target network (CIDR), domain name, and optional AD credentials. It validates input formats and forces the script to run as root (sudo). Operation levels are selected via simple menus for Scanning, Enumeration and Exploitation (None, Basic, Intermediate, Advanced).

Example: Level selection menu (short):

```
[+] Enter target network (e.g. 192.168.1.0/24): 192.16.1.0/24
[✓] Accepted network: 192.16.1.0/24
[+] Enter target domain name (e.g. myDomain.local): myDomain.local
[✓] Accepted domain: myDomain.local
[+] Enter AD Username (e.g. user@domain, blank for anonymous): Administrator
[+] Enter AD Password (blank for anonymous):
```

```
[+] Choose level for Scanning:
    0) None
    1) Basic
    2) Intermediate
    3) Advanced
[+] Enter number (0-3) or name:
```

2. Scanning Mode

Depending on the selected scanning level the script runs an Nmap TCP scan (basic: -Pn; intermediate/advanced: -sS -p-). For advanced UDP scanning it calls masscan (if installed). All scan outputs are saved under ./scan_results with timestamps.

Example of scanning levels menu:

```
[+] Choose level for Scanning:
    0) None
    1) Basic
    2) Intermediate
    3) Advanced
[+] Enter number (0-3) or name: 1
```

3. Enumeration Mode

Enumeration runs progressively: basic discovery (DHCP/DNS probe, targeted -sV), intermediate (CME shares/users, selected NSE scripts), and advanced (Impacket samrdump for users/groups/policy). The script attempts to identify a Domain Controller IP to focus enumeration.

Example of enumeration levels menu:

```
[+] Choose level for Enumeration:
    0) None
    1) Basic
    2) Intermediate
    3) Advanced
[+] Enter number (0-3) or name: 2
```

4. Exploitation Mode

Exploitation has three levels: Basic (Nmap vuln scripts), Intermediate (CME password spraying using a wordlist), Advanced (Impacket GetNPUsers.py for AS-REP roasting and optional hashcat cracking). Each step checks for required tools and prerequisites.

Example of exploitation levels menu:

```
[+] Choose level for Exploitation:
    0) None
    1) Basic
    2) Intermediate
    3) Advanced
[+] Enter number (0-3) or name: 3
```

5. Results and Report Generation

The script collects outputs (nmap, masscan, enumeration and exploitation logs) and writes a Markdown report. It attempts to convert the Markdown into PDF using pandoc or wkhtmltopdf. If conversion fails, a Markdown file is saved for manual conversion.

Summary

This automated script is useful for classroom labs and controlled assessments. Do NOT run it against networks you do not own or have permission to test. Always obtain authorization before performing scanning or exploitation activities.

Credits

Classmates, ChatGPT, HackTheBox

Important file locations

Scan results directory	./scan_results/
Nmap TCP output	./scan_results/nmap_tcp_scan_TIMESTAMP.txt
Enumeration output	./scan_results/enumeration_TIMESTAMP.txt
Exploitation output	./scan_results/exploitation_TIMESTAMP.txt
Generated report (MD/PDF)	/tmp/AD_SCAN_REPORT_TIMESTAMP.md or .pdf

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